

Galois-Informed HRV Analysis

Implementation and Negative Validation

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Status markers

- [B]** *Proved* — algebraically proved.
- [K]** *Confirmed* — numerically verified.
- [S]** *Speculative* — open.
- [L]** *Literature evidence* — external primary source.

Abstract

Docs. 358 and 359 provide the algebraic justification for a Galois-informed approach to HRV frequency analysis. The present document implements this approach as a Python module and validates it on two public ECG datasets. The outcome is a *negative validation* with a clear structural reason.

Results:

1. **Module [K]**: `galois_hrv_preprocess.py` — complete pipeline: Galois projection, resolution filter $100\xi \approx 1.33\%$, HRV band limits as constants, orbit-type classification, Arnold tongue width.
2. **BIDMC-01 [L]**: ICU patient, 88 y, 8 min PPG/ECG, Pimentel et al. 2017. Result: 0/3 ratios 5-limit, LF/VLF not resolvable (too short). Structural reason: recording too short for VLF, DFA $\alpha \approx 0.5$ (white noise, fractal regulation lost).
3. **Fantasia [L]**: 5 young + 5 elderly healthy subjects, each ≈ 120 min at rest, spontaneous breathing, Gold et al. 2002. Result: LF peak CV 11–33 % $\gg 1.33\%$, 5-limit LF/HF 4/10 (40 %), no group difference young/elderly.
4. **Structural reason [B]**: Under spontaneous breathing, HF = respiratory rate (variable). LF/HF is the ratio of two independent control loops and not a fixed Galois ratio.
5. **Next step [S]**: Controlled breathing at 6/min (resonance-frequency protocol, Lehrer & Gevirtz 2014) with a Polar H10 chest strap.

1 Galois-HRV Preprocessing Module

`galois_hrv_preprocess.py` implements (Doc. 359 §5):

Galois grid 901 rational ratios p/q with prime factors $\subseteq \{2, 3, 5, 7, 11, 13\}$ up to denominator 60.

Resolution filter $\Delta f/f \geq 100\xi \approx 1.33\%$ (Farey $Q_c \approx 11$, Doc. 343).

HRV band limits VLF: $[0, \frac{1}{25}]$ Hz, LF: $[\frac{1}{25}, \frac{3}{20}]$ Hz, HF: $[\frac{3}{20}, \frac{2}{5}]$ Hz — 5-limit constants **[B]** (Doc. 359 Theorem A).

Galois projection LF/HF $\mapsto$ nearest p/q in the grid.

Orbit type Galois depth $k = \text{ord}_p(3)$ of the prime factors (Docs. 343, 358).

Stability Arnold tongue width $\approx 2/(p+q)$ (Doc. 328).

Pipeline `hrv_galois_pipeline(rr_ms)` — a single function call.

2 Negative Validation I: BIDMC-01

Dataset: BIDMC PPG and Respiration Dataset v1.0.0, Record 01 **[L]** [?]. 88 y, male, MICU, 8 min, 125 Hz PPG + respiration.

Result: 0/3 frequency ratios satisfy the 5-limit criterion (Welch and Lomb–Scargle). VLF not resolvable (recording length 8 min < 15 min minimum for VLF). DFA $\alpha_1 = 0.50$, $\alpha_2 = 0.45$ — white noise; in healthy subjects ≈ 1.0 .

Why:

1. Too short for VLF (< 15 min).
2. ICU patient: fractal regulation largely lost.
3. HF peak = measured respiratory rate (0.328 Hz vs. RESP channel 0.336 Hz).
4. Measurement uncertainty Welch vs. Lomb–Scargle: $\pm 2\% >$ FFGFT limit 1.33 %.

3 Negative Validation II: Fantasia

Dataset: PhysioNet Fantasia Database v1.0.0 **[L]** [?]. 5 young (f1y01–05, 21–34 y) + 5 elderly (f2o01–05, 68–85 y), each ≈ 120 min at rest, 250 Hz ECG, spontaneous breathing. Annotation files (.ecg), CC-BY 4.0.

Methods: wfdb annotations, Welch with 10-min windows, zero padding $\times 4$, parabolic peak interpolation, LF stability over 6×5 -min segments.

Rec	n_{RR}	LF Hz	CV %	LF/HF	p/q	k	5-Lim	Arnold
f1y01	8707	0.109	11.5	0.438	7/16	6	N	0.087
f1y02	6919	0.085	33.0	0.867	13/15	4	N	0.071
f1y03	7641	0.100	13.7	4.219	4	1	Y	0.400
f1y04	5232	0.078	30.9	1.060	35/33	6	N	0.029
f1y05	6943	0.068	16.5	2.993	3	1	Y	0.500
f2o01	7045	0.067	25.6	1.494	3/2	1	Y	0.400
f2o02	6327	0.052	14.4	0.864	45/52	4	N	0.021
f2o03	6478	0.055	22.0	0.841	21/25	6	N	0.043
f2o04	6849	0.065	32.2	0.719	28/39	6	N	0.030
f2o05	8334	0.049	11.4	1.873	15/8	4	Y	0.087

Statistical tests (Mann–Whitney U): All group differences young vs. elderly are n.s. ($p > 0.40$).

Findings:

- LF peak stability: 0/10 records stable ($CV > 1.33\%$).
- 5-limit LF/HF: 4/10 (40 %) — not significantly above chance expectation ($\approx 30\text{--}40\%$).
- No group difference young/elderly (and $n = 5$ would not suffice anyway).

4 Structural Reason and Consequence

Finding. [B] Under spontaneous breathing, HF = respiratory rate (variable, 0.15–0.40 Hz). LF/HF is the ratio of two independent physiological control loops (baroreceptor / respiration) and does not converge to a fixed Galois ratio. Under spontaneous breathing the Galois test is structurally untestable.

Next step [S]: Controlled breathing at 6 breaths/min (0.10 Hz = LF resonance frequency). HF and LF then coincide; the system may lock in (broad Arnold tongue) or it may not. This is the Lehrer–Gevirtz effect [L] [?], which is physiologically well established.

Recommended setup:

- Polar H10 chest strap (1 ms RR resolution, ≈ 90 €).
- 5 min acclimatisation, 10 min at 6/min with a metronome.
- Export via Elite HRV or Kubios HRV (CSV).
- Analysis with `galois_hrv_preprocess.py`.

5 Breathing-Rate Scan: Results after Artefact Correction (10 September 2026)

Protocol: Polar H10 ECG chest strap, 130 Hz, seated, each 5–6 min, metronome click = phase change (4 bpm $\rightarrow$ 2/min; 8 bpm $\rightarrow$ 4/min; 10 bpm $\rightarrow$ 5/min; 12 bpm $\rightarrow$ 6/min). R-peak detection with artefact filtering (median $\pm 30\%$).

Condition	Breaths/min	LF Hz	RMSSD ms	Artefacts
Spontaneous (supine)	12.6	0.073	33.6	0
4/min	4.0	0.105	33.6	0
2/min Rec. 1	2.0	0.071	27.7	2
2/min Rec. 2	2.0	0.075	30.5	0
5/min Rec. 1	5.1	0.144	29.5	2
5/min Rec. 2	5.0	0.132	39.9	0
6/min	6.0	0.040	36.1	0

Finding 1 — No RMSSD resonance effect: After artefact cleaning, RMSSD lies between 27 and 40 ms in all conditions — no significant difference. Without artefact filtering, an apparent RMSSD maximum appears at 5/min (67.2 ms); it is an R-peak detection artefact (2 missed beats caused by amplitude modulation of the ECG during slow breathing, $|\Delta RR|_{\max} = 662$ ms).

Finding 2 — The LF peak moves with the breathing rate: The LF peak shifts with the breathing rate (0.040–0.144 Hz), but shows no stable resonance in the classical sense

($n = 1$). At 2/min the LF peak lies close to the spontaneous value (0.073 Hz) — at very slow breathing the baroreceptor eigenfrequency dominates.

Galois: LF/HF hits the grid at 5/min (88/35, 0.02 %), 6/min (39/35, 0.06 %) and spontaneous (56/45, 0.14 %) — all with $k = 6$. In no condition does LF coincide with the breathing rate.

Methodological lessons: Artefact filtering is mandatory under controlled breathing. Short recordings ($n = 1$) permit no statistical conclusions — all findings are exploratory.

6 Validation III: Polar H10 Self-Test (10 September 2026)

Device: Polar H10 ECG chest strap, firmware 4.2.0, JSONL export via Polar Sensor Recording (Android). Spontaneous breathing, supine, 21.9 min, 1517 RR intervals.

Parameter	Value	Assessment
HR	69.2 bpm	Normal at rest
RMSSD	36.3 ms	Good for 75 y (Fantasia elderly: 20–35 ms)
SDNN	53.8 ms	Normal
Breathing rate	12.6/min	Spontaneous
LF peak	0.073 Hz	Below norm (≈ 0.10 Hz)
LF/HF	1.2427	→ Galois 56/45, distance 0.14 %
LF CV	10.55 %	≫ 1.33 % (unstable)

Finding: Confirms the Fantasia negative validation. The LF CV of 10.55 % lies within the range of the Fantasia subjects (11–33 %). LF/HF lies in the Galois grid (distance 0.14 %), but with orbit depth $k = 6$ (prime 7, GF(729), R115) and low Arnold stability (0.020).

Positive finding: An RMSSD of 36.3 ms is very good for 75 y — clearly above the mean of the Fantasia “elderly” group (20–35 ms). Data quality of the Polar H10: flawless.

Note on the individual resonance frequency: The LF peak at 0.073 Hz (below the norm of 0.10 Hz) suggests that the individual resonance frequency may lie at 4–5/min rather than at 6/min. The individual protocol (testing rates of 4–8/min step by step) would clarify this — Lehrer & Gevirtz (2014) describe the determination procedure.

7 Summary

Result	Statement	Status
Module	galois_hrv_preprocess.py works	[K]
BIDMC-01	Negative: too short, ICU, no Galois signal	[L]
Fantasia	Negative: spontaneous breathing, LF CV 11–33 %	[L]
H10 self-test	Negative: LF CV 10.55 %, confirms Fantasia	[K]
Breathing scan	No RMSSD resonance effect after artefact correction	[K]
Galois	LF/HF at 5/min: 88/35 (0.02 %), $k = 6$	[K]
Reason	HF = respiration; LF moves; no coincidence of LF and breathing rate	[B]
Next	Longer recordings (≥ 15 min), $n > 5$	[S]

References

- [Pim17] Pimentel et al. (2017). BIDMC PPG and Respiration Dataset. PhysioNet. <https://physionet.org/content/bidmc/1.0.0/>
- [Gol02] Gold et al. (2002). Fantasia Database. PhysioNet. <https://physionet.org/content/fantasia/1.0.0/>
- [Leh14] Lehrer & Gevirtz (2014). Heart rate variability biofeedback. *Front. Psychol.* 5:756.
- [TF96] Task Force ESC/NASPE (1996). Heart rate variability. *Eur. Heart J.* 17, 354–381.

Scripts

2/python/Dok360_Skripte/:
fantasia_galois_analyse.py, galois_hrv_preprocess.py,
01_rpeaks_bidmc01.py, 02_welch_5limit.py, orbit_experiment_synthetic.py,
03_lombscargle_dfa.py. Fantasia
data: f1y01–05, f2o01–05 (.ecg/.hea, CC-BY 4.0).

Note on the use of AI assistance

AI assistance (Claude, Anthropic) was used in preparing this document for wording, LaTeX typesetting and analysis scripts. The substantive statements and the responsibility for the text rest with the author.